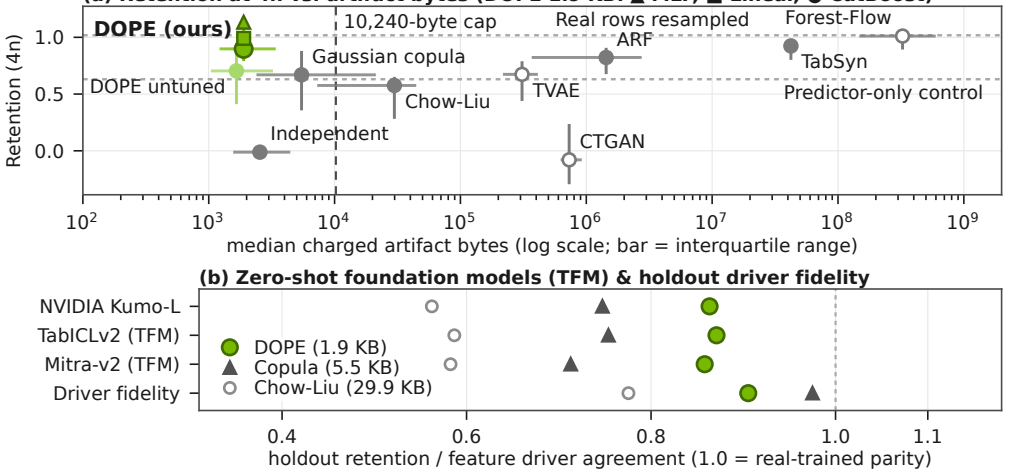
(a) Retention at 4n vs. artifact bytes (DOPE 1.9 KB: ▲ MLP, ■ Linear, ● CatBoost)
Retention (4n)
1.0
0.5
0.0
DOPE (ours)
10,240-byte cap
Real rows resampled
Forest-Flow
ARF
TabSyn
Gaussian copula
DOPE untuned
Chow-Liu
TVAE
Predictor-only control
Independent
CTGAN
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
median charged artifact bytes (log scale; bar = interquartile range)
(b) Zero-shot foundation models (TFM) & holdout driver fidelity
NVIDIA Kumo-L
TabICLv2 (TFM)
Mitra-v2 (TFM)
Driver fidelity
DOPE (1.9 KB)
Copula (5.5 KB)
Chow-Liu (29.9 KB)
0.4
0.6
0.8
1.0
1.1
holdout retention / feature driver agreement (1.0 = real-trained parity)